

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE COURSE [PHASE-III]

TEST # 06

DATE : 14-10-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
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MINOR TEST # 06

Topics

DATE : 14 - 10 - 2018

PRE-MEDICAL NURTURE COURSE [PHASE-III] : [NEET]

PHYSICS : Friction, Circular Motion, Work, Power & Energy

CHEMISTRY : Chemical Bonding, Gaseous State

BIOLOGY : Protista and Fungi Kingdom [Botany], Echinodermata to Agnatha [Zoology]

Next Up Coming Test

MINOR TEST # 07

Topics

DATE : 03 - 11 - 2018

PRE-MEDICAL NURTURE COURSE [PHASE-III] : [AIIMS]

PHYSICS : Work, Power, Energy, Circular Motion, Center of Mass, Collision, Rotational Motion

CHEMISTRY : Periodic Table, Atomic Structure, Chemical Bonding, Thermodynamics,
 Chemical Equilibrium

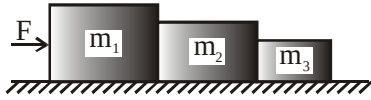
BIOLOGY : Animal Kingdom and Cockroach [Botany]
 Kingdom Fungi and Plant [Zoology]

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

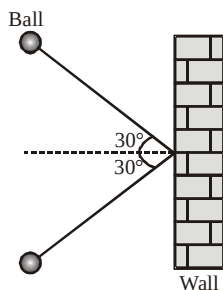
1. Three blocks of masses m_1 , m_2 and m_3 kg are placed in contact with each other on a frictionless table. A force F is



applied on the heaviest mass m_1 ; the acceleration of m_3 will be:

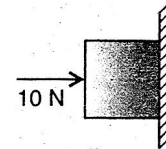
- (1) $\frac{F}{m_1}$ (2) $\frac{F}{m_1 + m_2}$
(3) $\frac{F}{m_2 + m_3}$ (4) $\frac{F}{m_1 + m_2 + m_3}$

2. A ball of mass 3 kg, moving with a speed of 100 m/s, strikes a wall at an angle 30° (as shown in figure). The ball rebounds at the same speed and remains in contact with the wall for 0.2 sec; the force exerted by the ball on the wall is :-



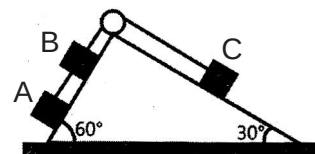
- (1) $1500\sqrt{3}$ N (2) 1500 N
(3) $300\sqrt{3}$ N (4) 300 N

3. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is:



- (1) 2 N
(2) 20 N
(3) 50 N
(4) 100 N

4. In the adjacent Fig., masses of A, B and C are 1 kg, 3 kg and 2 kg, respectively. Find
(a) the acceleration of the system and
(b) tension in the string A-B. Neglect friction
(Between A & B) ($g = 10\text{m/s}^2$)



- (1) 3.1 m/s, 2 N
(2) 4.1 m/s, 18.2 N
(3) 2.1 m/s, 9.2 N
(4) 3.1 m/s, 10.2 N

SPACE FOR ROUGH WORK

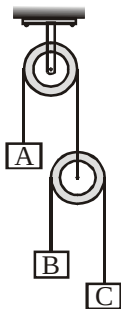
5. A rope of length L is pulled by a constant force F . What is the tension in the rope at a distance x from one end where the force is applied?

(1) $\frac{FL}{L-x}$ (2) $T = \frac{F(L-x)}{L}$
(3) Fx (4) $\frac{FL}{x}$

6. The trajectory of a projectile in a vertical plane is $y = ax - bx^2$, where a and b are constants and x and y are, respectively, horizontal and vertical distances of the projectile from the point of projection. The maximum height attained by the particle and the angle of projection from the horizontal are :-

(1) $\frac{b^2}{2a}, \tan^{-1}(b)$ (2) $\frac{a^2}{b}, \tan^{-1}(2b)$
(3) $\frac{a^2}{4b}, \tan^{-1}(a)$ (4) $\frac{2a^2}{b}, \tan^{-1}(a)$

7. If acceleration of block B is 4m/s^2 upward and that of C is 6m/s^2 downward. Find acceleration of A :-



- (1) 2 m/s^2 upward
(2) 1 m/s^2 upward
(3) 1 m/s^2 downward
(4) 2 m/s^2 downward

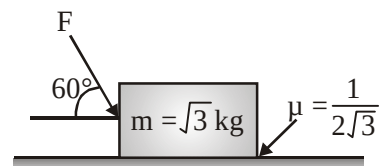
8. A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time t is proportional to :-

(1) $t^{3/4}$
(2) $t^{3/2}$
(3) $t^{1/4}$
(4) $t^{1/2}$

9. A body of mass 2 kg is kept stationary by pressing to a vertical wall by a force of 100 N . The coefficient of friction between wall and body is 0.3 then the frictional force is equal to

(1) 6 N (2) 20 N
(3) 600 N (4) 700 N

10. What is the maximum value of the force F such that the block shown in the arrangement of Fig. does not move ?



(1) 10 N (2) 20 N
(3) 30 N (4) 40 N

11. A particle starts from rest and moves with constant acceleration then ratio of distance travelled by it in 4th sec and in 5th sec is :-

(1) $\frac{16}{25}$ (2) $\frac{7}{9}$
(3) $\frac{7}{25}$ (4) $\frac{16}{9}$

SPACE FOR ROUGH WORK

12. If action force is gravitational force, then reaction force :-

- (1) May be electromagnetic force
- (2) May be electric force
- (3) May be spring force
- (4) Must be gravitational force

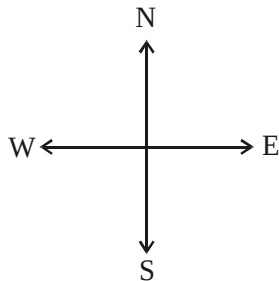
13. A physical quantity P is given by

$$P = \frac{A^3 \sqrt{B}}{C^2}, \text{ and the percentage errors in}$$

the measurements of A , B and C are 1%, 2% and 4% respectively. Find the percentage error in P .

- (1) 10
- (2) 12
- (3) 13
- (4) 14

14. Two body A and B are 10 km apart such that B is in south of A . A and B start moving with same speed 20 km/hr eastward and northward respectively then find :-



- (a) Relative velocity of A w.r.t. B
- (b) Minimum separation attained during motion
- (c) Time lapse, from starting to attain, minimum separation

(1) $21\sqrt{2} \frac{\text{km}}{\text{hr}}$ B , $11 \times \frac{1}{\sqrt{2}}$ km, 14 min.

(2) $20\sqrt{2} \frac{\text{km}}{\text{hr}}$ B , $10 \times \frac{1}{\sqrt{2}}$ km, 15 min.

(3) $22\sqrt{2} \frac{\text{km}}{\text{hr}}$ B , $12 \times \frac{1}{\sqrt{2}}$ km, 16 min.

(4) $23\sqrt{2} \frac{\text{km}}{\text{hr}}$ B , $11 \times \frac{1}{\sqrt{2}}$ km, 18 min.

15. The water drops fall at regular intervals from a tap 5 m above the ground. The third drop is leaving the tap at the instant the first drop touches the ground. How far above the ground is the second drop at that instant?

- (1) 1.25 m
- (2) 2.50 m
- (3) 3.75 m
- (4) 4.00 m

16. Initial velocity of a particle is v_0 and its acceleration is $a = kt$ is a constant. Calculate velocity.

(1) $v_0 + \frac{k}{2}t^2$

(2) $v_0 - \frac{k}{2}t^2$

(3) $\frac{2v_0}{k}$

(4) $\frac{2v_0}{k} + t^2$

SPACE FOR ROUGH WORK

17. A goods train acceleration uniformly on a straight railway track, approaches an electric pole standing on the side of track. Its engine passes the pole with velocity u and the guard's room passes with velocity v . The middle wagon of the train passes the pole with a velocity :-

(1) $\sqrt{\left(\frac{u^2 + v^2}{2}\right)}$ (2) $\frac{1}{2}\sqrt{u^2 + v^2}$

(3) $\sqrt{uv}$ (4) $\frac{u+v}{2}$

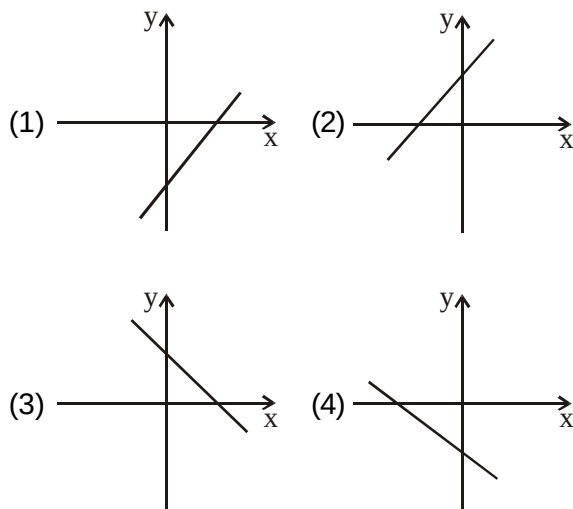
18. Differentiate y w.r.t x

$$y = \frac{\tan x}{x}$$

(1) $x^2 \cos x$ (2) $\frac{x \tan x - \cos x}{x^2}$

(3) $\frac{x \sec^2 x - \tan x}{x^2}$ (4) $\frac{-x \cos x + \sin x}{x^2}$

19. Which of following graph is the best representation for equation $y = 2x - 4$:-



20. A physical quantity P is related to four observables a, b, c and d as follows; $P = \frac{a^3 b^2}{\sqrt{c} d}$.

The percentage errors of measurement in a, b, c and d are 1%, 3%, 4% and 2%, respectively. What is the percentage error in the quantity P ?

(1) 10 (2) 11

(3) 12 (4) 13

21. A particle is projected with a velocity v , so that its range on a horizontal plane is twice the greatest height attained. If g is acceleration due to gravity, then its range is :-

(1) $\frac{4v^2}{5g}$ (2) $\frac{4g}{5v^2}$

(3) $\frac{4v^3}{5g^2}$ (4) $\frac{4v}{5g^2}$

22. The equation of a projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$.

The angle of projection is and initial velocity is

(1) $30^\circ, 2$ (2) $45^\circ, 1$

(3) $45^\circ, 2$ (4) $60^\circ, 2$

23. A stone is dropped from the top of a tower and travels 24.5 m in the last second of its journey. The height of the tower is :-

(1) 44.1m (2) 49m

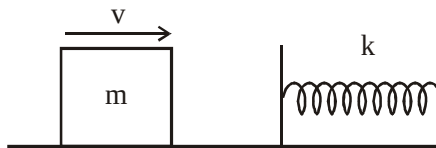
(3) 78.4m (4) 72m

SPACE FOR ROUGH WORK

24. The potential energy of a spring when stretched through a distance x is 10 J. What is the amount of work done on the same spring to stretch it through an additional distance x ?

(1) 10 J (2) 20 J
(3) 30 J (4) 40 J

25. A mass of 0.5 kg moving with a speed of 1.5 m/s on a horizontal smooth surface collides with a nearly weightless spring of force constant $k = 50$ N/m. The maximum compression of the spring would be :



(1) 0.15 m (2) 0.12 m
(3) 1.5 m (4) 0.5 m

26. A position dependent force $F = 7 - 2x + 3x^2$ acts on a small body of mass 2 kg and displaces it from $x = 0$ to $x = 5$ m. Calculate the work done in joule.

(1) 135 J (2) 137 J
(3) 139 J (4) 133 J

27. A stone of mass 1 kg tied to a light string

length $\ell = \frac{10}{3}$ m is whirling in a circular path in vertical plane. If the ratio of the maximum to minimum tension in the string is 4, find the speed of the stone at the lowest and highest points :-

(1) $v_1 = 15.2 \text{ ms}^{-1}$, $v_2 = 10 \text{ ms}^{-1}$
(2) $v_1 = 14.2 \text{ ms}^{-1}$, $v_2 = 12 \text{ ms}^{-1}$
(3) $v_1 = 13.2 \text{ ms}^{-1}$, $v_2 = 13 \text{ ms}^{-1}$
(4) $v_1 = 16.2 \text{ ms}^{-1}$, $v_2 = 11 \text{ ms}^{-1}$

28. The P.E. of a particle oscillating on x-axis is given as $U = 20 + (x - 2)^2$ here U is in Joules & x is in meters. Total mechanical energy of particle is 36 J

(i) Find the mean position
(ii) Find the max. K.E. of the particle

(1) (i) $x = 2$, (ii) 16 J
(2) (i) $x = 3$, (ii) 15 J
(3) (i) $x = 4$, (ii) 17 J
(4) (i) $x = 1$, (ii) 10 J

29. A force of magnitude of 30 N acting along $\hat{i} + \hat{j} + \hat{k}$, displaces a particle from point (2, 4, 1) to (3, 5, 2). The work done during this displacement is :-

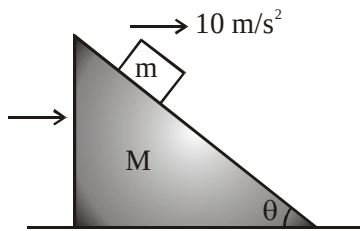
(1) 90 J
(2) 30 J
(3) $30\sqrt{3}$ J
(4) $30/\sqrt{3}$ J

30. A spring of force constant 800 N/m has an extension of 5 cm. The work done in extending it from 5 cm to 15 cm is :-

(1) 16 J
(2) 8 J
(3) 32 J
(4) 24 J

SPACE FOR ROUGH WORK

31. In the figure shown all the surfaces are frictionless, and mass of the block, $m = 1$ kg. The block and wedge are held initially at rest. Now wedge is given a horizontal acceleration of 10 m/s^2 by applying a force on the wedge, so that the block does not slip on the wedge. Then work done by the normal force in ground frame on the block in $\sqrt{3}$ seconds is :-

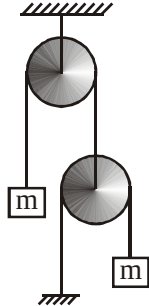


- (1) 30 J (2) 60 J
(3) 150 J (4) $100\sqrt{3}$ J
32. A rope is used to lower vertically a block of mass M by a distance x with a constant downward acceleration $g/2$. The work done by the rope on the block is :-
- (1) Mgx (2) $\frac{1}{2}Mgx^2$
(3) $-\frac{1}{2}Mgx$ (4) Mgx^2
33. A body of mass 0.5 kg travels in a straight line with velocity $v = ax^{3/2}$ where $a = 5 \text{ m}^{-1/2}\text{s}^{-1}$. The work done by the net force during its displacement from $x = 0$ to $x = 2$ m is :-
- (1) 1.5 J (2) 50 J
(3) 10 J (4) 100 J

34. A man who is running has half the kinetic energy of the boy of half his mass. The man speeds up by 1 m/s and then has the same kinetic energy as the boy. The original speed of the man was :-
- (1) $\sqrt{2} \text{ m/s}$
(2) $(\sqrt{2} - 1) \text{ m/s}$
(3) 2 m/s
(4) $(\sqrt{2} + 1) \text{ m/s}$
35. A mass of 5 kg is moving along a circular path of radius 1 m. If the mass moves with 300 revolutions per minutes, its kinetic energy would be :-
- (1) $250 \pi^2$ joule
(2) $100 \pi^2$ joule
(3) $5 \pi^2$ joule
(4) 0 joule
36. A particle moves on a rough horizontal ground with some initial velocity say v_0 . If $3/4$ of its kinetic energy is lost in friction in time t_0 then coefficient of friction between the particle and the ground is :-
- (1) $\frac{v_0}{2gt_0}$
(2) $\frac{v_0}{4gt_0}$
(3) $\frac{3v_0}{4gt_0}$
(4) $\frac{v_0}{gt_0}$

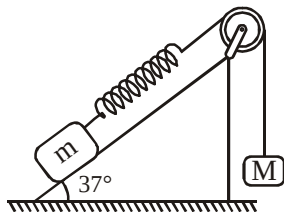
SPACE FOR ROUGH WORK

37. In the figure shown, the system released from rest. Find the velocity of block A when block B has fallen a distance ' ℓ '. Assume all pulleys to be massless and frictionless :-



- (1) $\sqrt{\frac{g\ell}{5}}$ (2) $\sqrt{g\ell}$
(3) $\sqrt{5g\ell}$ (4) None of these

38. A block of mass m is attached with a massless spring of force constant k . The block is placed over a rough inclined surface for which the coefficient of friction is $\mu = \frac{3}{4}$. The minimum value of M required to move the block up the plane is : (Neglect mass of string and pulley and friction in pulley)



- (1) $\frac{3}{5}m$ (2) $\frac{4}{5}m$
(3) $2m$ (4) $\frac{3}{2}m$

39. A body of mass m accelerates uniformly from rest to v_1 in time t_1 . The instantaneous power delivered to the body as a function of time t is :-

- (1) $\frac{mv_1 t}{t_1}$ (2) $\frac{mv_1^2 t}{t_1^2}$
(3) $\frac{mv_1 t}{t_1}$ (4) $\frac{mv_1^2 t}{t_1}$

40. A light spring of length 20 cm and force constant 2 N/cm is placed vertically on a table. A small block of mass 1 kg falls on it. The length h from the surface of the table at which the block will have the maximum velocity is :-

- (1) 20 cm
(2) 15 cm
(3) 10 cm
(4) 5 cm

41. In a shotput event an athlete throws the shotput of mass 10 kg with an initial speed of 1 m/s at 45° from a height 1.5 m above ground. Assuming air resistance to be negligible and acceleration due to gravity to be 10 m/s^2 , the kinetic energy of the shotput when it just reaches the ground will be :-

- (1) 2.5 J
(2) 5.0 J
(3) 52.5 J
(4) 155.0 J

SPACE FOR ROUGH WORK

42. The potential energy of a 1 kg particle free to move along the x-axis is given by

$$V(x) = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \text{J. The total mechanical}$$

energy of the particle is 2 J. Then, the maximum speed (in m/s) is :-

(1) $3/\sqrt{2}$

(2) $\sqrt{2}$

(3) $1/\sqrt{2}$

(4) 2

43. A 2 kg block slides on a horizontal floor with a speed of 4 m/s. It strikes an uncompressed spring, and compresses it till the block is motionless. The kinetic friction force is 15 N and spring constant is 10, 000 N/m. The spring compresses by :-

(1) 5.5 cm

(2) 2.5 cm

(3) 11.0 cm

(4) 8.5 cm

44. When a rubber-band is stretched by a distance x, it exerts a restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is :-

(1) $\frac{aL^2}{2} + \frac{bL^2}{3}$

(2) $\frac{1}{2} \left(\frac{aL^2}{2} + \frac{bL^2}{3} \right)$

(3) $aL^2 + bL^3$

(4) $\frac{1}{2}(aL^2 + bL^3)$

45. A body of mass $m = 10^{-2}$ kg is moving in a medium and experiences a frictional force $F = -kv^2$. Its initial speed is $v_0 = 10 \text{ ms}^{-1}$. If,

after 10 s, its energy is $\frac{1}{8}mv_0^2$, the value of

k will be :-

(1) $10^{-4} \text{ kg m}^{-1}$

(2) $10^{-1} \text{ kg m}^{-1} \text{ s}^{-1}$

(3) $10^{-3} \text{ kg m}^{-1}$

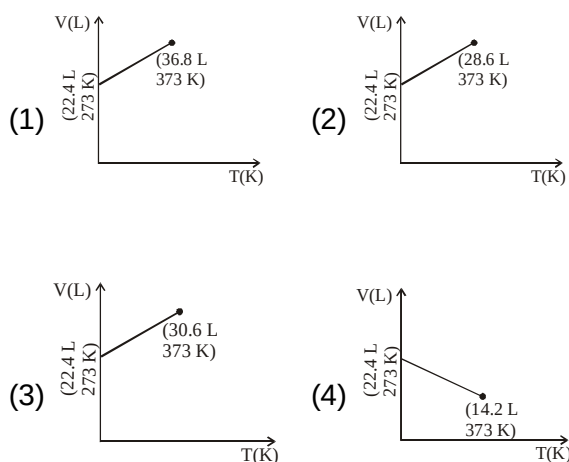
(4) $10^{-3} \text{ kg s}^{-1}$

SPACE FOR ROUGH WORK

- | | |
|---|--|
| <p>46. The maximum number of molecules are present in :</p> <p>(1) 15 L of H_2 gas at STP</p> <p>(2) 5 L of N_2 gas at STP</p> <p>(3) 0.5g of H_2 gas</p> <p>(4) 10 g of O_2 gas</p> <p>47. Insulin contains 3.4% sulphur. Then, the minimum molecular mass of the insulin is about:</p> <p>(1) 940 amu</p> <p>(2) 9400 amu</p> <p>(3) 3600 amu</p> <p>(4) 970 amu</p> <p>48. 25 g of MCl_4 contains 0.5 mol chlorine then its molecular mass is:</p> <p>(1) 100 g mol^{-1}</p> <p>(2) 200 g mol^{-1}</p> <p>(3) 150 g mol^{-1}</p> <p>(4) 400 g mol^{-1}</p> <p>49. Which of the following has $p\pi - d\pi$ bonding?</p> <p>(1) NO_3^- (2) SO_3^{2-}</p> <p>(3) BO_3^{3-} (4) CO_3^{2-}</p> <p>50. Atomic mass of chlorine is 35.5. It has two isotopes of atomic mass 35 and 37. The percentage of heavier isotope is:</p> <p>(1) 10 (2) 15</p> <p>(3) 20 (4) 25</p> <p>51. The vapour densities of two gases are in the ratio of 1 : 3. Their molecular masses are in the ratio of:</p> <p>(1) 1 : 3 (2) 1 : 2</p> <p>(3) 2 : 3 (4) 3 : 1</p> | <p>52. The atomic mass of an element is 27. If valency is 3, the vapour density of the volatile chloride will be:</p> <p>(1) 66.75</p> <p>(2) 6.675</p> <p>(3) 667.5</p> <p>(4) 81</p> <p>53. The density of a gas 'A' is three times that of a gas 'B'. If the molecular mass of A is M, the molecular mass of B is:</p> <p>(1) 3M</p> <p>(2) $M/3$</p> <p>(3) $\sqrt{3}M$</p> <p>(4) $\frac{M}{\sqrt{3}}$</p> <p>54. The largest number of molecules is in:</p> <p>(1) 28 g of CO</p> <p>(2) 46 g of C_2H_5OH</p> <p>(3) 36 g of H_2O</p> <p>(4) 54 g of N_2O_5</p> <p>55. How many moles of magnesium phosphate, $Mg_3(PO_4)_2$, will contain 0.25 mole of oxygen atoms?</p> <p>(1) 0.02</p> <p>(2) 3.125×10^{-2}</p> <p>(3) 1.25×10^{-2}</p> <p>(4) 2.5×10^{-2}</p> |
|---|--|

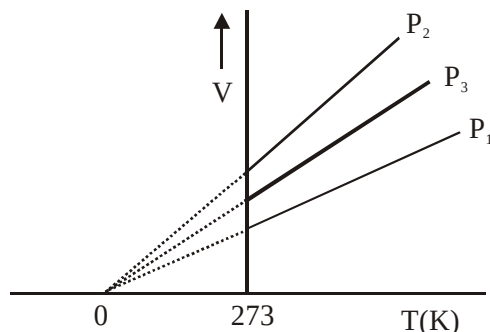
SPACE FOR ROUGH WORK

56. Which of the following volume (V)–temperature (T) plots represents the behaviour of one mole of an ideal gas at one atmospheric pressure ?



57. The volume temperature graphs of a given mass of an ideal gas at constant pressure are shown below :

What is the correct order of pressure?



- (1) $P_1 > P_3 > P_2$
(2) $P_1 > P_2 > P_3$
(3) $P_2 > P_3 > P_1$
(4) $P_2 > P_1 > P_3$

58. Select one correct statement. In the gas equation, $PV=nRT$

- (1) n is the number of molecules of a gas
(2) V denotes volume of one mole of the gas
(3) n moles of the gas have a volume V
(4) P is the pressure of the gas when only one mole of gas is present.

59. A gaseous mixture of 2 moles of A, 3 moles of B, 5 moles of C and 10 moles of D is contained in a vessel. Assuming that the gases are ideal and the partial pressure of C is 1.5 atm, the total pressure is

- (1) 3 atm (2) 6 atm
(3) 9 atm (4) 15 atm

60. The ratio of rates of diffusion of SO_2 , O_2 and CH_4 is

- (1) $1:\sqrt{2}:2$ (2) $1:2:4$
(3) $2:\sqrt{2}:1$ (4) $1:2:\sqrt{2}$

61. According to Graham's law, at a given temperature, the ratio of the rates of diffusion r_A/r_B of gases A and B is given by (where p and M are pressures and molecular weights of gases A and B)

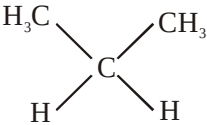
(1) $\left(\frac{p_A}{p_B}\right)\left(\frac{M_A}{M_B}\right)^{1/2}$

(2) $\frac{M_A}{M_B}\left(\frac{p_A}{p_B}\right)^{1/2}$

(3) $\left(\frac{p_A}{p_B}\right)\left(\frac{M_B}{M_A}\right)^{1/2}$

(4) $\frac{M_A}{M_B}\left(\frac{p_B}{p_A}\right)^{1/2}$

SPACE FOR ROUGH WORK

62. Helium atom is two times heavier than a hydrogen molecule at 298 K. The average kinetic energy of helium is
 (1) two times that of a hydrogen molecule
 (2) same as that of a hydrogen molecule
 (3) four times that of a hydrogen molecule
 (4) half that of a hydrogen molecule
63. For gaseous state, if most probable speed is denoted by C^* , average speed by $\bar{C}$ and root mean square speed by C , then for a large number of molecules the ratios of these speeds are
 (1) $C^* : \bar{C} : C = 1 : 1.128 : 1.225$
 (2) $C^* : \bar{C} : C = 1 : 1.225 : 1.128$
 (3) $C^* : \bar{C} : C = 1.225 : 1.128 : 1$
 (4) $C^* : \bar{C} : C = 1.128 : 1.225 : 1$
64. For H_2 gas, the compressibility factor, $Z = PV/nRT$ is :
 (1) equal to 1
 (2) equal to 0
 (3) always greater than 1
 (4) initially less than 1 and then becomes greater than 1 at high pressures
65. The Vander Waal's constant 'a' for the gases O_2 , N_2 , NH_3 and CH_4 are 1.3, 1.390, 4.170 and 2.253 $L^2 \text{ atm mol}^{-2}$ respectively. The gas which can be most easily liquefied is
 (1) O_2 (2) N_2 (3) NH_3 (4) CH_4
66. The compressibility factor of a gas is less than unity at STP. Therefore,
 (1) $V_m > 22.4 \text{ L}$ (2) $V_m < 22.4 \text{ L}$
 (3) $V_m = 22.4 \text{ L}$ (4) $V_m = 44.8 \text{ L}$
67. The relationship between P_c , V_c and T_c is
 (1) $P_c V_c = RT$
 (2) $P_c V_c = 3RT_c$
 (3) $P_c V_c = \frac{3}{5}RT_c$
 (4) $P_c V_c = \frac{3}{8}RT_c$
68. The correct order of bond length in the following is :-
 (1) $CO^+ < CO < CO_2 < CO_3^{2-}$
 (2) $CO < CO^+ < CO_2 < CO_3^{2-}$
 (3) $CO_2 < CO < CO^+ < CO_3^{2-}$
 (4) $CO_3^{2-} < CO < CO_2 < CO^+$
69. Which of the following is an example of super octet molecule ?
 (1) ClF_3 (2) PCl_5
 (3) IF_7 (4) All the three.
70. Which of the following hydrocarbons has the lowest dipole moment ?
 (1) 
 (2) $CH_3-C \equiv C-CH_3$
 (3) $CH_3CH_2C \equiv CH$
 (4) $CH_2 = CH-C \equiv CH$
71. In an ionic compound A^+X^- , the degree of covalent bonding is greatest when
 (1) A^+ and X^- ions are small
 (2) A^+ is small and X^- is large
 (3) A^+ and X^- ions are approximately of the same size
 (4) X^- is small and A^+ is large.

SPACE FOR ROUGH WORK

72. The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^{2+} , Mg^{2+} , Ba^{2+} ?

 - $Ca^{2+} < Mg^{2+} < Be^{2+} < K^+$
 - $Mg^{2+} < Be^{2+} < K^+ < Ca^{2+}$
 - $Be^{2+} < K^+ < Ca^{2+} < Mg^{2+}$
 - $K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$

73. A molecule MX_3 is T-shaped. The number of non-bonding pairs of electrons is

 - 0
 - 2
 - 1
 - can be predicted only if atomic number of M is known.

74. In OF_2 , number of bond pairs and lone pairs of electrons are respectively

 - 2, 6
 - 2, 8
 - 2, 10
 - 2, 9

75. In XeF_2 , XeF_4 and XeF_6 , the number of lone pairs on Xe is respectively

 - 2, 3, 1
 - 1, 2, 3
 - 4, 1, 2
 - 3, 2, 1

76. There are five pairs of bonding electrons and one pair of non-bonding electrons around the central atom of a molecule. The expected geometry and shape of the molecule will be

 - tetrahedral, planar
 - octahedral, planar
 - octahedral, square pyramidal
 - trigonal bipyramidal, square planar

77. In a change from $PCl_3 \longrightarrow PCl_5$, the hybrid state of P changes from

 - sp^2 to sp^3
 - sp^3 to sp^2
 - sp^3 to sp^3d
 - sp^3 to dsp^2 .

78. The shape of ClF_3 molecule is

 - Triangular planar
 - Pyramidal
 - T-shape
 - Trigonal bipyramidal

79. Of the following hydrides, which one has the lowest boiling point?

 - AsH_3
 - SbH_3
 - PH_3
 - NH_3

80. Stability of the species Li_2 , Li_2^- and Li_2^+ increases in the order of

 - $Li_2 < Li_2^- < Li_2^+$
 - $Li_2^- < Li_2 < Li_2^+$
 - $Li_2 < Li_2^+ < Li_2^-$
 - $Li_2^- < Li_2^+ < Li_2$

81. The species C_2

 - has one σ bond and one π bond
 - has both π bonds
 - has both σ bonds
 - does not exist.

82. In which of the following pairs of molecule/ions, both the species are not likely to exist ?

 - H_2^{2+} , He_2
 - H_2^- , He_2^{2+}
 - H_2^+ , He_2^-
 - H_2^{2-} , He_2^{2-}

83. The covalent bond length is the shortest in which one of the following bonds ?

 - $C-O$
 - $C \equiv N$
 - $C \equiv C$
 - $O-H$

SPACE FOR ROUGH WORK

84. Which of the following bonds is the strongest ?

- (1) F—F
- (2) I—I
- (3) Cl—Cl
- (4) Br—Br.

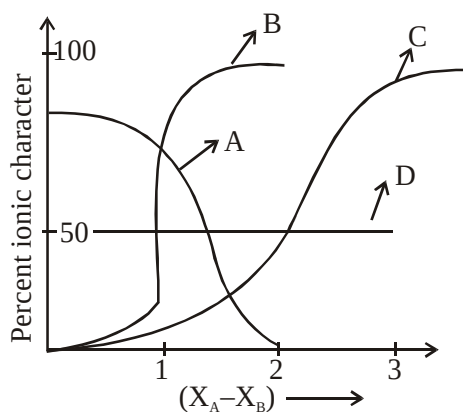
85. The highest bond energy is for the bond

- (1) N—N
- (2) F—F
- (3) C—C
- (4) O—O

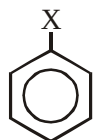
86. Among the following, the molecule with highest dipole moment is

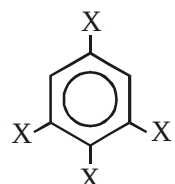
- (1) CH₃Cl
- (2) CH₂Cl₂
- (3) CHCl₃
- (4) CCl₄

87. For AB bond if percent ionic character is plotted against electronegativity difference ($X_A - X_B$), the shape of the curve would look like :-



- (1) (A)
- (2) (B)
- (3) (C)
- (4) (D)

88. The dipole moment of  is 1.5 D.

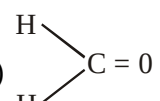
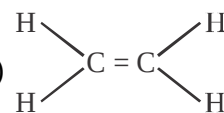
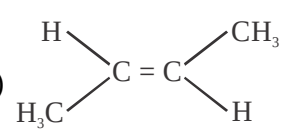
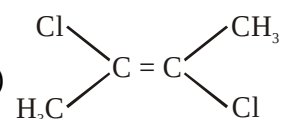
The dipole moment of  is :-

- (1) 1.5 D
- (2) 2.25 D
- (3) 1 D
- (4) 3 D

89. The correct order of increasing covalent character of the following is :-

- (1) SiCl₄ < AlCl₃ < CaCl₂ < KCl
- (2) KCl < CaCl₂ < AlCl₃ < SiCl₄
- (3) AlCl₃ < CaCl₂ < KCl < SiCl₄
- (4) none of these

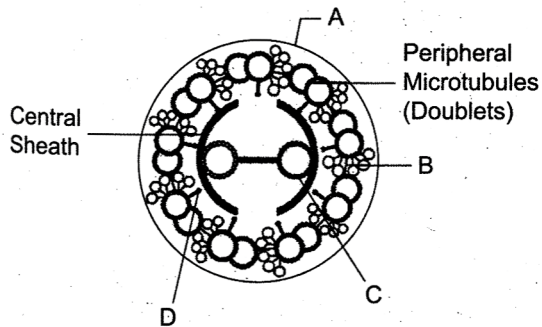
90. Which has highest dipole moment?

- (1) 
- (2) 
- (3) 
- (4) 

SPACE FOR ROUGH WORK

91. Unicellular organism are :-
 (1) not capable of independent existence because they cannot perform all the essential functions of life
 (2) not capable of independent existence but they can perform all the essential vital functions.
 (3) are capable of independent existence and perform all the essential vital functions
 (4) are capable of independent existence but they perform some vital function
92. Which of the following is not a characteristic of prokaryotic cell?
 (1) Mesosome
 (2) Circular DNA
 (3) Photosynthetic membrane system
 (4) Membrane bound organelles
93. Select the incorrect statement.
 (1) Filament of the bacterial flagella is the longest part and extends from the cell surface to the outside
 (2) In prokaryotes, ribosomes (70S) are associated with plasma membrane
 (3) The size of prokaryotic ribosomes is $15 \text{ nm} \times 20 \text{ nm}$
 (4) None of these
94. The best material for the study of structure of cell membrane is
 (1) RBC of human beings
 (2) RBC of frog
 (3) Kidney cell
 (4) Muscle cell
95. According to the modern concept, cell membrane is
 (1) solid
 (2) quasi fluid
 (3) fluid
 (4) solidified sheath
96. The innermost portion of a mature plant cell wall is the
 (1) primary cell wall
 (2) plasma membrane
 (3) secondary cell wall
 (4) plasmodesmata
97. Which of the following is the correct sequence/ route of the secretory product?
 (1) ER $\rightarrow$ Vesicles $\rightarrow$ Cis region of GB $\rightarrow$ Trans region of GB $\rightarrow$ Vesicle $\rightarrow$ Plasma membrane
 (2) RER $\rightarrow$ GB $\rightarrow$ Lysosome $\rightarrow$ Nuclear membrane $\rightarrow$ Plasma membrane
 (3) ER $\rightarrow$ Vesicles $\rightarrow$ Trans region of GB $\rightarrow$ Cis region of GB $\rightarrow$ Vesicle $\rightarrow$ Plasma membrane
 (4) Lysosome $\rightarrow$ ER $\rightarrow$ GB $\rightarrow$ Vesicles $\rightarrow$ Cell membrane
98. The concentration of a number of ions and other materials is higher in vacuoles than those in cytoplasm, why?
 (1) Tonoplast has a number of active transport system that pumps ions into vacuoles from the cytoplasm
 (2) Through osmosis, a large amount of ions go continuously to vacuole from the cytoplasm
 (3) Cytoplasmic ions enter the vacuole through osmotic flow of water
 (4) Vacuole is always engaged in the hydrolysis of salts into their ions
99. Section of Cilia/flagella shows
- | | Peripheral Microtubules (Double) | Central microtubules (singlet) | Radial spoke | Central Sheath |
|-----|----------------------------------|--------------------------------|--------------|----------------|
| (1) | 9+0 | 2 | 8 | 1 |
| (2) | 9+2 | 9+0 | 9 | 1 |
| (3) | 9 | 2 | 9 | 1 |
| (4) | 3 | 6 | 9 | 1 |

100. Go through the section of cilia/flagella showing the different parts as follows :



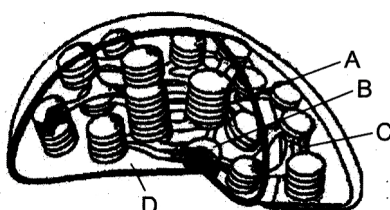
In which of the following options correct words for all the four blanks A, B, C and D are given?

- (1) A-Plasma membrane, B-Interdoublet bridge, C-Central microtubule, D-Radial spoke
- (2) A-Plasma membrane, B-Arm, C-Central microtubule, D-Radial spoke
- (3) A-Plasma membrane, B-Interdoublet bridge, C-Hub, D-Radial spoke
- (4) A-Plasma membrane, B-Interdoublet bridge, C-Hub, D-Arm

101. The crosssection of a centriole shows :-

	Peripheral Microtubules (Triplet)	Central Microtubules (Singlet)	Hub	Spokes	Inter-Triplet Bridge
(1)	9	2	1	9	9
(2)	9	2	9	9	9
(3)	9	2	1	2	2
(4)	9	0	1	9	9

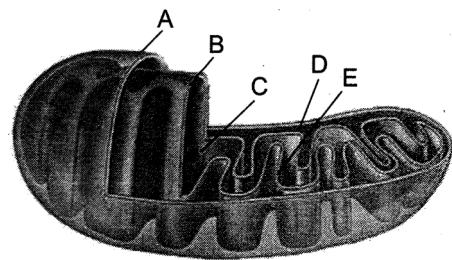
102. Go through the section view of chloroplast showing the different parts.



In which of the following options, correct words for all the four blanks A, B, C, and D are indicated

- (1) A-Thylakoid, B-Stromal lamella, C-Stroma, D-Granum
- (2) A-Granum, B-Thylakoid, C-Stromal lamella, D-Stroma
- (3) A-Thylakoid, B-Granum, C-Stromal lamella, D-Stroma
- (4) A-Granum, B-Thylakoid, C-Stroma, D-Stromal lamella

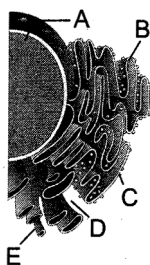
103. Go through the section view of a mitochondrion



Now, identify A to E.

- (1) A-Outer membrane, B-Inner membrane, C-Matrix, D-Inter-membrane space, E-Crista
- (2) A-Outer membrane, B-Inner membrane, C-Inter-membrane space, D-Matrix, E-Crista
- (3) A-Outer membrane, B-Inner membrane, C-Matrix, D-Crista, E-Inter-membrane space
- (4) A-Outer membrane, B-Inner membrane, C-Crista, D-Matrix, E-Inter-membrane space

104. Identify the components labelled A, B, C, D and E in the diagram below from the list (i) to (viii) given.



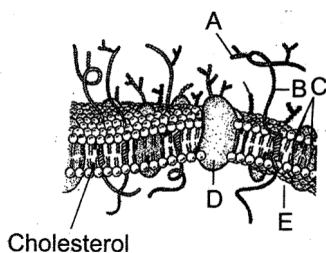
Components :

- (i) Cristae of mitochondria
- (ii) Inner membrane of mitochondria
- (iii) Cytoplasm
- (iv) Smooth endoplasmic reticulum
- (v) Rough endoplasmic reticulum
- (vi) Mitochondrial matrix
- (vii) Ribosome
- (viii) Nucleus

The correct matches are

- (1) A-(viii) B-(v) C-(vii) D-(iii) E-(iv)
- (2) A-(i) B-(iv) C-(vii) D-(vi) E-(iii)
- (3) A-(vi) B-(v) C-(iv) D-(vii) E-(i)
- (4) A-(v) B-(i) C-(iii) D-(ii) E-(iv)

105. Identify the components labelled A, B, C, D and E in the diagram (cell membrane) below from the list (i) to (viii) given.



- (i) Sugar
- (ii) Protein
- (iii) Lipid bilayer
- (iv) Integral protein
- (v) Cytoplasm
- (vi) Cell wall
- (vii) External protein

The correct components are

- (1) A-(i) B-(ii) C-(iii) D-(iv) E-(v)
- (2) A-(ii) B-(i) C-(iii) D-(iv) E-(v)
- (3) A-(i) B-(ii) C-(iii) D-(iv) E-(vi)
- (4) A-(i) B-(ii) C-(iii) D-(vii) E-(v)

106. At metaphase of mitosis, each chromosome consists of _____ chromatid(s) _____ centromere(s) kinetochore(s) and _____ molecules of DNA.

- (1) 2, 1, 2, 2 (2) 2, 2, 2, 2
- (3) 2, 1, 1, 1 (4) 2, 1, 2, 1

107. The events shown below occur during different phases.

- (A) Centromere splits, chromatids separate and move to opposite poles. Chromatids are now called chromosomes
- (B) Chromosomes cluster at opposite poles, decondensation of chromosome reappearance of NM, GB, ER and nucleolus
- (C) Chromosomal replication
- (D) Kinetochore attach to spindle fibres and chromosome are arranged at equatorial plate
- (E) Condensation of chromosomal materials

Which of the following correctly identifies each of the phases described? Interphase Prophase Metaphase Anaphase Telophase

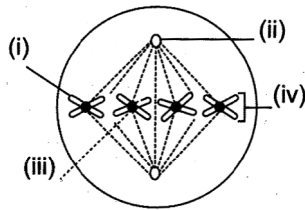
- (1) C E D A B (2) C D E A B
- (3) C E D B A (4) C A D E B

108. Mother Cell

Identify A to C.

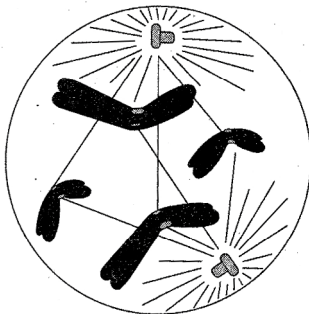
- | | A | B | C |
|-------------|---|----|----|
| (1) Meiosis | | N | N |
| (2) Meiosis | | 2N | 2N |
| (3) Mitosis | | N | N |
| (4) Mitosis | | 2N | 2N |

109. Label the structure indicated by lines (i), (ii), (iii) and (iv).



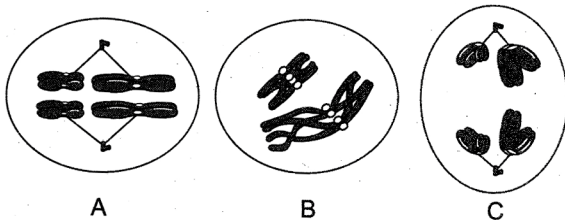
- (1) (i) Chromatid, (ii) Centriole, (iii) Centromere, (iv) Chromosome
- (2) (i) Chromosome, (ii) Centriole, (iii) Centromere, (iv) Chromatid
- (3) (i) Chromatid, (ii) Centromere, (iii) Centriole, (iv) Chromosome
- (4) (i) Chromosome, (ii) Centromere, (iii) Centriole, (iv) Chromatid

110. Identify the following diagram.



- (1) Transition to Metaphase (mitosis)
- (2) Transition to Anaphase II (meiosis)
- (3) Transition to Metaphase I (meiosis)
- (4) Transition to Anaphase (mitosis)

111. Certain stages of cell division are represented by given figures. Identify them correctly.



- (1) A-Metaphase, B-Prophase I, C-Anaphase
- (2) A-Metaphase, B-Prophase I, C-Anaphase I
- (3) A-Metaphase I, B-Prophase I, C-Anaphase I
- (4) A-Metaphase I, B-Prophase, C-Anaphase

112. How many equational divisions are necessary in a cell of onion root tip to form 128 cells?

- (1) 64
- (2) 128
- (3) 7
- (4) None of these

113. Which of the following features are generally characteristics of life?

- I. Growth
- II. Reproduction
- III. Response to stimuli
- IV. Metabolism
- V. Cellular organization

- (1) I, II, IV, V
- (2) All of these
- (3) II, III, IV
- (4) I, II, III, V

114. A group of real plants or animals with similar traits of any rank is

- (1) species
- (2) genus
- (3) order
- (4) taxon

115. Linnaeus system of classification was based on

- (1) morphology
- (2) ecology
- (3) embryology
- (4) cytology

116. The sequence of taxonomic categories is

- (1) Class → Phylum → Tribe → Order → Family → Genus → Species
- (2) Division → Class → Family → Tribe → Order → Genus → Species
- (3) Division → Class → Order → Family → Tribe → Genus → Species
- (4) Phylum → Order → Class → Tribe → Family → Genus → Species

117. Which one of the following has least similar characters?

- (1) Family
- (2) Class
- (3) Genus
- (4) Species

118. On which of the following, the five-kingdom classification is based?

- (1) Cell wall
- (2) Genetic material
- (3) Types of organelles
- (4) Mode of nutrition

119. Which of the following statements about classification is not true ?

- I. Members of a family are less similar than members of an included genus
- II. An order has more members than the number of members in an included genus
- III. Families have more members than phyla
- IV. Members of a family share a common ancestor in the more distant past than members of an included genus
- V. The number of species in a taxon depends on their relative degree of similarity

- (1) Only III (2) Only IV
(3) Only V (4) None of these

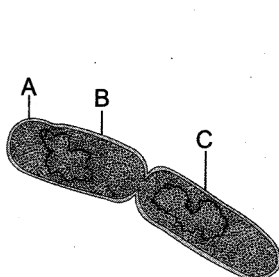
120. ICBN is

- (1) International Code of Biological Naming
- (2) International Code of Botanical Nomenclature
- (3) Internal Class of Biological Nomenclature
- (4) International Classification of Biological Nomenclature

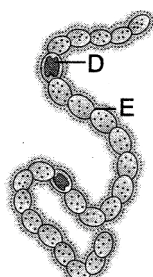
121. Which of the following is self-conscious ?

- (1) Human being (2) Tiger
(3) Lion (3) Amoeba

122. Identify the labellings in the following figures.



A Dividing Bacteria



Nostoc

- (1) A-Cell wall, B-Cell membrane, C-Heterocyst, D-NA, E-Mucilaginous sheath
- (2) A-Cell wall, B-Cell membrane, C-DNA, D-Heterocyst, E-Mucilaginous sheath

- (3) A-Mucilaginous sheath, B-Cell membrane, C-DNA, D-Heterocyst, E-Cell wall
- (4) A-Cell membrane, B-Cell wall, C-DNA, D-Heterocyst, E-Mucilaginous sheath

123. Which of the following kingdoms has no well defined boundaries?

- (1) Monera
- (2) Protista
- (3) Fungi
- (4) Metaphyta and Metazoa

124. In which of the following, the cell wall has stiff cellulose plate on the outer surface?

- (1) Dinoflagellates
- (2) Desmids
- (3) Diatoms
- (4) Euglenoids

125. Read the following statements carefully and select correct option :

- (a) In zoological parks, conditions similar to natural habitats are provided to animals.
- (b) Keys are generally analytical in nature.
- (c) In herbarium sheet, local names are not mentioned.
- (d) Taxonomical aids are useful in knowing bioresources and their diversity.

- (1) Only (a) and (b)
- (2) (a), (c) and (d)
- (3) (a), (b) and (d)
- (4) All of these

126. Select the incorrect statements :

- (1) Museums often have collections of skeletons of animals.
- (2) Separate taxonomic keys are required for each taxonomic category.
- (3) Taxonomic keys are based on the contrasting characters.
- (4) Herbarium is a store house of collected living plant specimens.

127. Select the correct combination of the statements regarding the characters of certain organisms :
- Methanogens are archaebacteria which produce methane in marshy areas.
 - Chemosynthetic autotrophic bacteria synthesise cellulose from glucose.
 - Mycoplasma lack of cell wall and survive in presence of oxygen.
 - Nostoc perform N_2 fixation in heterocyst cell of filament
- a and b
 - a and d
 - b and c
 - c and d
128. Which statements are right for euglena :
- They have unequal flagella.
 - They represent only autotrophic mode of nutrition.
 - They are found in fresh water.
 - They have pigments similar to angiosperms.
 - They may obtain food by predating on other small organisms.
- b, c & e
 - a, b & d
 - a, b, c & e
 - a, c, d & e
129. Match the following Column I and Column II :
- | Column I | Column II |
|---------------------|---------------------------------------|
| (1) Slime mold | (i) Diatoms |
| (2) Dinoflagellates | (ii) Yellow and blue colour |
| (3) Chrysophytes | (iii) Gonyaulax |
| | (iv) Saprophytic |
| | (v) Represent soap box like structure |
| | (vi) Fruiting body formation |
- a - ii, iii b - iv, vi c - i, v
 - a - iv, vi b - ii, iii c - i, v
 - a - iii, vi b - i, iv c - ii, v
 - a - v, vi b - i, iv c - ii, iii
130. How many given characters of organisms are used as main criteria for five kingdom classification :
- Cell structure
 - Thallus organisation
 - Reproduction
 - Mode of nutrition
 - Venation of leaf in plants
 - Phylogenetic relationship
 - Pigments of leaf perform photosynthesis.
- 4
 - 5
 - 2
 - 3
131. A pair of bacteria involved in two steps during oxidation of ammonia into nitrate :
- Clostridium and Azotobacter
 - Nitrosomonas and nitrobacter
 - Rhizobium and Pseudomonas
 - Xanthomonas and Rhodospirillum
132. Which of the following is incorrect statement :-
- Entire DNA of donor cell enter into the recipient cell during transduction of bacteria
 - Transduction is method of genetic recombination in some bacteria
 - Conjugation tube in bacterial conjugation sexual reproduction prepared by Pili
 - Extra chromosomal DNA in bacteria called plasmid
133. Consider the following taxonomic aids
- Botanical garden
 - Museum
 - Zoological park
 - Herbarium
 - Key
- How many are helpful in taxonomic studies of plants and animals :-
- 4
 - 1
 - 5
 - 2
134. Select wrong statement for Heterocyst :-
- They are found in *Nostoc* and *Anabaena*
 - They are found in *Chlorella*
 - They have nitrogenase enzyme to perform N_2 fixation
 - Mostly cyanobacteria found in water of paddy field and perform N_2 fixation in heterocyst
135. Most common method of reproduction in bacteria during favourable condition is :-
- Spore formation
 - Conjugation
 - Budding
 - Binary fission

136. Read the following five statements [A-E] :-

- (A) It forms the lining of the cavities of alveoli of lungs
- (B) It forms the lining of wet surface like buccal cavity and oesophagus
- (C) It occurs in the ducts of sweat glands
- (D) It forms the lining of salivary glands and sweat glands
- (E) It is a loose connective tissue

Which of the above are associated with simple epithelial tissue ?

- (1) A and D
- (2) B and C
- (3) C and A
- (4) D and E

137. Pavement epithelium is another name of:

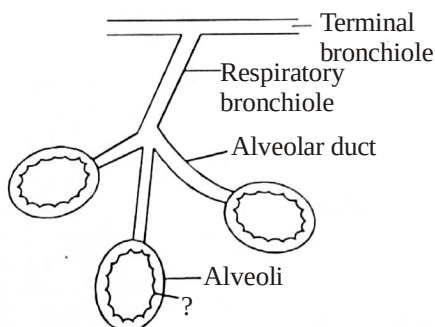
- (1) Stratified epithelium
- (2) Ciliated epithelium
- (3) Simple squamous epithelium
- (4) Cuboidal epithelium

138. Match the following columns :-

Column-I		Column-II	
A.	Areolar tissue	1.	Fat cells
B.	Adipose tissue	2.	Osteocytes
C.	Ligament	3.	Loose connective tissue
D.	Bone	4.	Dense regular C.T.

- (1) A-3, B-1, C-4, D-2
- (2) A-1, B-2, C-3, D-4
- (3) A-4, B-3, C-2, D-1
- (4) A-2, B-1, C-4, D-3

139. Identify the type of epithelium tissue and their function

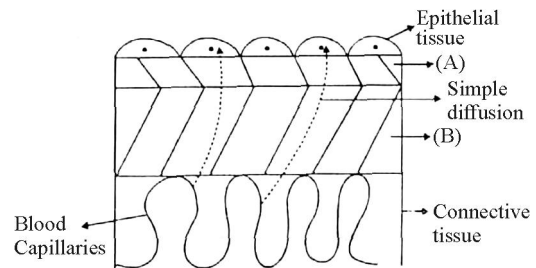


- (1) Simple cuboidal epithelium = absorption
- (2) Simple columnar epithelium = secretion
- (3) Simple squamous epithelium = diffusion
- (4) Pseudostratified epithelium = filtration

140. Urinary bladder, renal pelvis, ureter and proximal part of male urethra consist of which type of compound epithelium ?

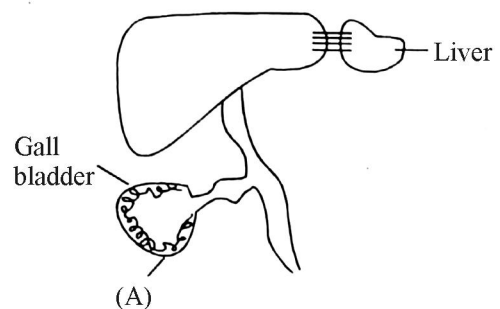
- (1) Simple columnar epithelium
- (2) Stratified squamous epithelium
- (3) Transitional epithelium
- (4) Ciliated columnar epithelium

141. In the following diagram (A) and (B) parts are made up of following composition :-



- (1) (A) = Glycoprotein
(B) = Mucopolysaccharide
- (2) (A) = Hyaluronic acid
(B) = Keratin protein
- (3) (A) = Scleroprotein
(B) = Hyaluronic acid
- (4) (A) = Collagen fibres
(B) = Glycoprotein

142. In given diagram identify the type of tissue



- (1) Ciliated columnar epithelium
- (2) Steriociliated columnar epithelium
- (3) Brush border columnar epithelium
- (4) Glandular columnar epithelium

143. Match list I with list II and find the correct option:-

List I		List II (Location)	
(A)	Cuboidal	(1)	Epidermis of skin
(B)	Ciliated	(2)	Inner lining of blood
(C)	Columnar	(3)	Inner surface of gall bladder
(D)	Squamous	(4)	Inner lining of Fallopian tube
(E)	Keratinized squamous	(5)	Lining of pancreatic duct

(1) (A)-(5), (B)-(4), (C)-(2), (D)-(3), (E)-(1)

(2) (A)-(3), (B)-(4), (C)-(5), (D)-(2), (E)-(1)

(3) (A)-(5), (B)-(4), (C)-(3), (D)-(2), (E)-(1)

(4) (A)-(3), (B)-(4), (C)-(5), (D)-(1), (E)-(2)

144. Macrophage cell is a connective tissue cell, the Function of which is :-

(1) Phagocytic

(2) Secretion

(3) Antibody production

(4) Fibre production

145. Yellow fibrous connective tissue is found in

(a) Renal capsule

(b) True vocal cords

(c) Walls of blood vessels

(d) Epimysium

(1) a, b (2) b, c

(3) c, d (4) a, b, c

146. Tall and slender cells are present in :-

(1) Squamous epithelium

(2) Columnar epithelium

(3) Pseudostratified epithelium

(4) Both (2) and (3)

147. Endoneurium, Epimysium, Perimysium, Endomysium, respectively are :-

(1) Areolar C.T, Sheath of WFCT, sheath of WFCT, Areolar C.T.

(2) Sheath of WFCT, Areolar C.T., Areolar C.T, Areolar C.T.

(3) Areolar C.T, Sheath of WFCT, Areolar C.T, Areolar C.T.

(4) Areolar C.T., Cord of WFCT, Areolar C.T., Areolar C.T.

148. Read the following statements (A-D)

(A) Muscular tissue is most abundant and widely distributed tissue.

(B) All connective tissue cells secrete fibres of structural proteins called collagen or elastin.

(C) Epithelial cells rest on basement membrane.

(D) Blood often serves as support framework of epithelium.

How many above statements are incorrect?

(1) One

(2) Three

(3) Two

(4) Four

149. Which one is incorrect match :-

(1)	Compound epithelium	Main role in secretion and absorption
(2)	Areolar tissue	Support frame work of epithelium
(3)	Blood	Specialised C.T.
(4)	Dense regular C.T.	Tendon and ligaments

150. Semifluid ground substance is most abundant in ?

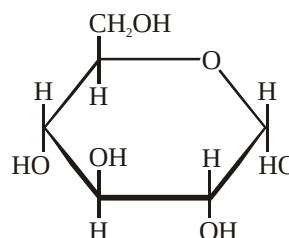
(1) Dense regular connective tissue

(2) Areolar connective tissue

(3) All specialised connective tissue

(4) Simple columnar epithelium

151. Identify the given structure and name the compound.



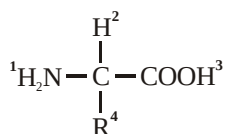
(1) Ribose

(2) Sucrose

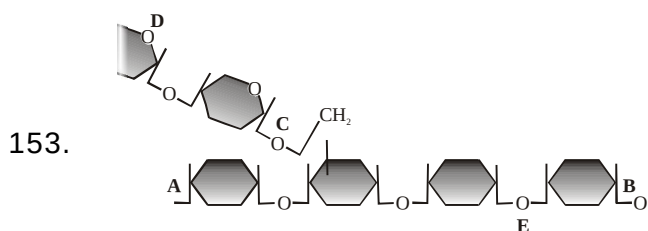
(3) Glucose

(4) Ribulose

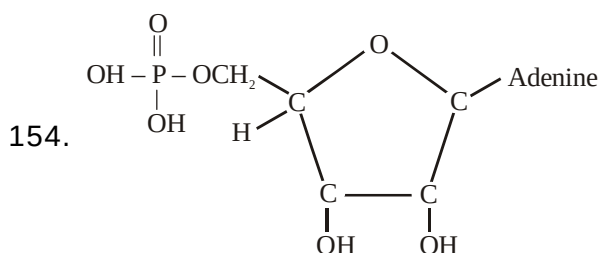
152. Which of the two groups of following formula is involved in peptide bond formation between different amino acids?



- (1) 1 and 3 (2) 2 and 3
(3) 2 and 4 (4) 1 and 4

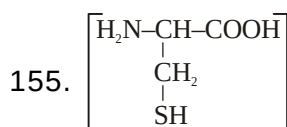


- | Column-I | Column-II |
|----------|-----------------------|
| (A) | (i) reducing end |
| (B) | (ii) non reducing end |
| (C) | (iii) α 1 – 6 |
| (D) | (iv) α 1 – 4 |
| (E) | (v) β 1 – 4 |
- (1) A - i, B - ii, C - iv, D - i, E - iii
 (2) A - ii, B - i, C - v, D - i, E - iii
 (3) A - ii, B - i, C - iii, D - ii, E - iv
 (4) A - i, B - ii, C - iii, D - i, E - iv



Above given structure represent :-

- (1) Nucleotide of DNA
- (2) Nucleotide of RNA
- (3) Nucleoside of DNA
- (4) Nucleoside of RNA



What is the name of given amino acid?

- (1) Serine
- (2) Tyrosine
- (3) Cysteine
- (4) proline

156. Zwitter-ionic form of amino acid is :-

- $$\begin{array}{ll} \text{(1) } \text{H}_3\text{N}-\underset{\text{R}}{\text{CH}}-\text{COOH} & \text{(2) } \text{H}_3\text{N}^+-\underset{\text{R}}{\text{CH}}-\text{COO}^- \\ \text{(3) } \text{H}_2\text{N}-\underset{\text{R}}{\text{CH}}-\text{COO}^- & \text{(4) } \text{H}_2\text{N}^+-\underset{\text{R}}{\text{CH}}-\text{COOH} \end{array}$$

157. Find out incorrect matched pair?

- (1) Pigment – Anthocyanine, Carotenoid
- (2) Toxin – Morphine, Codeine
- (3) Lectin – Concanavalin A
- (4) Polymeric substances – Gum, Rubber

158. If water = 1, Protein = 2, Carbohydrate = 3, Lipid = 4, Nucleic acid = 5 then which of the following is correct order of their % weight of total cellular mass in our body?

- (1) $1 > 2 > 4 > 3 > 5$
 (2) $1 > 2 > 5 > 3 > 4$
 (3) $2 > 1 > 3 > 5 > 4$
 (4) $1 > 2 > 3 > 4 > 5$

159. Match the column and choose correct combination :-

Column-I		Column-II	
(a)	Acidic amino acid	(i)	Valine
(b)	Basic amino acid	(ii)	Glutamic acid
(c)	Neutral amino acid	(iii)	Phenyl alanine
(d)	Aromatic amino acid	(iv)	Lysine

- | a | b | c | d |
|---------|----|-----|-----|
| (1) i | ii | iv | iii |
| (2) ii | i | iii | iv |
| (3) ii | iv | i | iii |
| (4) iii | ii | iv | i |

160. Which of the following statement is not correct regarding amino acid?

- (1) The structure of amino acid changes in solution of different pHs.
- (2) COOH group of amino acid having ionizable nature
- (3) NH₂ group of amino acid having ionizable nature
- (4) Amino acid found in protein belongs to D-forms mostly.

161. Find out incorrect statement?

- (1) Lipids are micromolecules
- (2) All monosaccharides occurs in d and l form except dihydroxy acetone
- (3) C₄ epimer of glucose is also known as blood sugar
- (4) Fructose is also known as Laevulose since it rotate PPL into left direction

162. Match the following column and find out incorrect match?

Disaccharide	Bond Direction	Nature
(1) Maltose	α 1 $\rightarrow$ 4	Reducing
(2) Sucrose	$\alpha\beta$ 1 $\leftrightarrow$ 2	Non-reducing
(3) Trehalose	β 1 $\leftrightarrow$ 2	Non-reducing
(4) Lactose	β 1 $\rightarrow$ 4	Reducing

163. Glycogen, Starch, Pectin, Hemicellulose, Inulin, Chondroitin, Chitin are given numbers 1, 2, 3, 4, 5, 6, 7 respectively on the basis of given information find out correct option?

Homopolysaccharide
Heteropolysaccharide

- | | |
|----------------|------------|
| (1) 1, 2, 5, 7 | 3, 4, 6 |
| (2) 2, 3, 6 | 1, 4, 5, 7 |
| (3) 1, 2, 4, 6 | 3, 5, 7 |
| (4) 3, 5, 6 | 1, 2, 4, 7 |

164. Find out incorrect statement from given below :-

- (1) Inulin is a polymer of fructose
- (2) In a polysaccharide chain right end is called the reducing end and the left and is called the non-reducing end.
- (3) Starch forms helical structure and it can hold I₂ molecule
- (4) Collagen is most abundant protein in the whole of the biosphere

165. MUFA and PUFA are ____ (a) ____ fatty acid and metabolically ____ (b) ____ reactive molecules 'a' and 'b' is respectively?

- (1) Saturated, Highly
- (2) Essential, Less
- (3) Essential, Highly
- (4) Non-essential, Less

166. The "chagas" disease is caused by-

- (1) *Leishmania donovani*
- (2) *Leishmania tropica*
- (3) *Trypanosoma gambiensi*
- (4) *Trypanosoma cruzi*

167. The classification of phylum-protista is based on-

- (1) Locomotion
- (2) Respiration
- (3) Nutrition
- (4) Reproduction

168. The vector for African sleeping sickness is-

- (1) *Glossina palpalis*
- (2) *Phlebotomus* fly
- (3) *Tabanus* fly
- (4) Bed-bug

169. Which of the following sponge has radial symmetry?

- | | |
|-------------------------|--------------------|
| (1) <i>Spongilla</i> | (2) <i>Cliona</i> |
| (3) <i>Leucosolenia</i> | (4) <i>Spongia</i> |

170. Select the mismatch pair-

- (1) Scleroblast - Formation of skeleton
- (2) Thesocytes - Regulate opening and closing of osculum
- (3) Chromocytes - Pigmented cell
- (4) Trophocytes - Distribution of food

171. Select the incorrect match-
- (1) Obelia - Sea fur
 - (2) Pennatula - Sea fan
 - (3) Sea hare - Aplysia
 - (4) Sea lily - Echinus
172. Which of the following is not a major phylum?
- (1) Cnidaria
 - (2) Ctenophora
 - (3) Mollusca
 - (4) Annelida
173. Choose the phylum in which all the animals are triploblastic and acoelomate-
- (1) Platyhelminthes
 - (2) Aschelminthes
 - (3) Annelida
 - (4) Mollusca
174. Coelom formed by splitting of mesodermal mass is/are found in-
- (1) Annelida
 - (2) Arthropoda
 - (3) Molluscs
 - (4) All
175. Animals in which anus is formed earlier than mouth
- (1) Annelida
 - (2) Arthropoda
 - (3) Mollusca
 - (4) Echinodermata
176. Choose the option which describes the Aschelminthes with its exclusive features
- (1) Rounded unsegmented body, muscular pharynx, pseudocoelomate, females are longer than males
 - (2) Rounded segmented body, muscular pharynx, pseudocoelomate, females are longer than males
 - (3) Rounded unsegmented body, muscular pharynx, acoelomate, females are longer than males
 - (4) Rounded unsegmented body, muscular pharynx, pseudocoelomate, females are smaller than males

177. Find out the incorrect match about the characters of annelida
- (1) Longitudinal and circular muscles help in locomotion
 - (2) Parapodia help in swimming
 - (3) Flame cells help in excretion
 - (4) Neural system consist of paired ganglia
178. Which of the following statement / character is incorrect about the diagram given above?



- (1) Posses lateral appendages, parapodia which help in reproduction
 - (2) Body surface distinctly marked out into segments
 - (3) Circulatory system is closed
 - (4) Nephridia help in excretion and osmoregulation
179. In which of the following the animal name, its two characters and its phylum / class are correctly matched?

Animal name		Two characters	Nature
(1)	Pila	Body segmented Mouth with radula	Mollusca
(2)	Asterias	Spiny skin Water canal system	Echinodermata
(3)	Periplanata	Jointed appandages Chitinous	Arthropoda
(4)	Saccoglossus	Post anal tail present	hemichordata

180. Identify the animals shown in the given figures A, B, C and D from the options given below :-



(A)



(B)



(C)



(D)

	A	B	C	D
(1)	Pila	Octopus	Asterias	Ophiura
(2)	Octopus	Pila	Asterias	Ophiura
(3)	Asterias	Ophiura	Pila	Octopus
(4)	Ophiura	Asterias	Pila	Octopus

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